

Encryption cuts the chance of a costly breach

Given

Expected annual loss
\$800k

P(breach | unencrypted) 0.40

P(breach | encrypted) 0.08

Loss per breach (ALE) \$2,000,000

Risk = probability x loss

Unencrypted: $0.40 \times 2,000,000 = \$800,000$

\$160k

Encrypted: $0.08 \times 2,000,000 = \$160,000$

unencrypted encrypted
Risk reduction = \$640,000 saved per year